

What Does a Computer-Use Agent Reliability Score Actually Measure?

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Abstract

A computer-use agent reliability score is treated as if it observed whether the agent did the task. It does not. The score is a constructed measurement: a pipeline that maps a trajectory into evidence, a decision, and a number. We audit one frozen instrument on MyPCBench with three complementary studies, not three independent experiments. Study 1 shows that the number need not track the world the agent saw: across 24 valid counterfactual pairs the score can stay put when tracking holds, and can move when tracking does not. Study 2 shows that the same number, used as a selection instrument, is coverage-limited and degenerate: three agents each ran 57 legs; valid pairs numbered 9, 8, and 1; binary tracking was 0 on every valid pair; the confirmatory selection test was, by its own rule, not evaluated. On the remaining four-task common support the score and a graded tracking layer disagree, but the disagreement is carried by one task. Study 3 opens the evaluator. Of 59 observations in which recoverable evidence was present, the frozen instrument missed 39; 13 of those misses were fail-closed aggregation discarding a gold value it had already collected. Layer-specific repairs do not dominate: recovering more evidence is not recovering the intended measurement, and a channel intervention inverts an ordering statistic by destroying one lane while leaving the other unchanged ($n_{\text{eff}} = 1$). A locked, corpus-independent reconstruction rule recovers 9 of 30 hand-written label components. Reliability is not read off the score. It is constructed, and correspondence can fail at identifiable stages.

CCS Concepts

• Computing methodologies → Intelligent agents; • General and reference → Evaluation.

Keywords

computer-use agents, reliability, evaluation, measurement validity

ACM Reference Format:

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1 Introduction

Leaderboards for computer-use agents (CUAs) report a completion-conditioned score: the episode ended, and a rubric assigned a number [3, 9, 10]. Selecting an agent from that number treats the score as an observation of reliability. The treatment is undefined. A score is a *constructed measurement* [2, 5]: a sequence of transformations that map a trajectory into candidate evidence, a decision, and a

reported value. Validity is a property of those transformations, not of the scalar they emit.

This paper asks one question: *what does a CUA reliability score actually measure?* We answer it on one frozen instrument, with three complementary studies that open the same measurement system at three resolutions (Figure 1). They are not three independent experiments: the benchmark family, the paired counterfactual protocol, and the model roster overlap. We do not add sample sizes across studies.

Study 1 asks whether the score tracks the world the agent saw. Paired counterfactual episodes hold the instruction, interface, and judge fixed while a locked intervention rewrites a pre-registered determining set. The score can remain unchanged across materially different outcomes, and can vary without a corresponding change in tracking.

Study 2 asks whether the score at least performs the selection job it is used for. After a coverage-first inclusion rule, only four task clusters support a pairwise comparison; binary tracking is identically zero on the valid-pair universe; the confirmatory selection test is not evaluated.

Study 3 asks where correspondence is lost. Recoverable evidence is frequently missed; fail-closed aggregation discards gold it has already collected; layer-specific repairs have signed, non-monotonic effects; and the hand-written specification is not reconstructed by a locked grounded rule.

Contributions. (1) A pipeline decomposition of CUA reliability evaluation, with world correspondence as the validity check rather than an extra score. (2) A motivating dissociation between score and outcome (Study 1) and a functional audit of the score as a selection instrument (Study 2). (3) A diagnosis of the evaluator: evidence loss, aggregation discard, non-monotonic repairs, an existence demonstration that a layer intervention can move a comparative statistic, and a failed reconstruction of the frozen specification (Study 3).

The intended audience is the community that builds and scores interactive agents. If a reliability number is used to pick a CUA, to retire a CUA, or to claim that an agent “did the task,” then the maps that produce the number are part of the scientific claim. An audit of those maps is an agent-evaluation result, not a complaint about one vendor judge.

We do not claim that CUA benchmarks are unreliable, that a repair should be shipped, or that rankings are unstable in general. A pre-specified comparative-scale gate required 20 independent task clusters; 16 of 28 candidates survived a frozen feasibility probe, so that experiment was not opened.

2 Reliability as a measurement pipeline

A CUA episode produces a trajectory τ in a guest world E . A reliability instrument J does not observe reliability. It applies a sequence of maps:

$\tau \mapsto \text{candidates} \mapsto \text{reported value} \mapsto S(\tau) \mapsto \text{selection}.$

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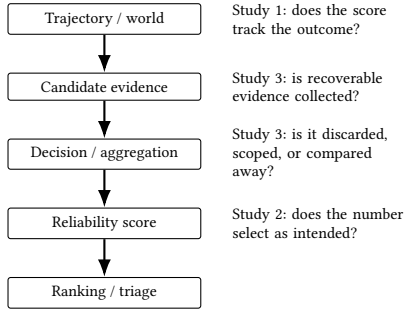


Figure 1: The score is constructed. Studies are lenses on the same pipeline, not consecutive experiments. World correspondence is the validity check, not a stage.

Correspondence can fail at any map. Study 1 tests the composite $S(\tau)$ against the world. Study 2 tests S as a selection instrument against a tracking construct that never consults J . Study 3 inspects the intermediate maps of J itself.

Shared protocol. All three studies use MyPCBench [3], a seeded personal desktop, and a paired counterfactual design [6]. For task t , E_t^0 is the seeded guest and $E_t^1 = I_t(E_t^0)$ applies a locked patch to a pre-registered determining set D_t . Instruction, interface, screenshot size, and judge stay fixed; guest probes confirm that gold moved. A pair is *valid* only if both legs terminate canonically. Completion is read from the last recorded action by a fail-closed classifier that does not consult the judge, so closure and score are disjoint channels. Execution failures are absent observations, not tracking misses.

Frozen instrument. Study 3 treats the existing evaluator as an object to be audited, not improved. Labels, windows, aggregation, and comparison rules are held at a recorded commit. Repairs in Study 3 are diagnostic interventions on one layer at a time; they are not a proposed replacement metric.

3 Study 1: Score–outcome dissociation

Does a reliability score necessarily track a change in the underlying outcome?

On each valid pair we read three facts separately. *Tracking* is a typed match of the final answer against guest gold on D , with no judge in the loop. *Score sensitivity* is exact equality $S_{CF} = S_{Base}$ versus any nonzero delta. A pair is Type A if tracking holds and the score is invariant; score-sensitive if tracking holds and the score moves; Type B if tracking is incomplete while the score is high and unchanged.

The design is a frozen 5×10 schedule (50 intended cells; 4 never scheduled; 46 executed; 92 legs; 35 execution failures; 24 valid pairs). Primary lanes are Claude, GPT-5.5, and Qwen3.5-35B-A3B. A size ablation and an exploratory Flash lane are not pooled into the rates below.

Claude contributes 9 valid pairs, of which 7 are tracking-valid: 6 Type A, 1 score-sensitive, 2 Type B; invariance $6/7 = 0.857$ (Clopper–Pearson 95% CI [0.421, 0.996]). GPT-5.5 contributes 8 valid, 7 tracking-valid: 3 Type A, 4 score-sensitive, 1 Type B; $3/7 =$

Table 1: Study 1 primary lanes. Invariance is Type A over tracking-valid pairs. Intervals are Clopper–Pearson 95%.

Lane	Valid	Track-valid	Type A	Sens.	Type B
Claude	9	7	6	1	2
GPT-5.5	8	7	3	4	1
Qwen3.5-35B-A3B	1	1	1	0	0

Table 2: Coverage under the frozen instrument. A valid pair requires both legs to terminate. $|\mathcal{A}|$ is a lower bound: the terminal classifier is fail-closed. Claude’s mean S^0 is a single pair.

Agent	DONE/57	$ \mathcal{A} $	mean S^0	mean STS
GPT-5.5	32	9	69.3	0.130
Qwen3.8-Flash	29	8	95.8	0.229
Claude Opus 4.6	4	1	100	0

0.429 ([0.099, 0.816]). Qwen3.5-35B-A3B contributes one valid pair, Type A ([0.025, 1.000]).

Tracking is coded from the final answer, never from S . Incomplete runs are execution failures. Intervals are wide (Claude [0.421, 0.996]; GPT [0.099, 0.816]; Qwen [0.025, 1.000]). We do not headline a pooled invariance rate, and we do not add these pairs to Study 2’s 18 valid pairs. A designed “trap” label does not determine the observed class: the same intervention can be a clean Type A on one agent and a split on another.

The claim is the phenomenon: the reported score can remain unchanged across materially different outcomes, and can vary without a corresponding change in tracking. A stable score does not certify tracking; a moving score does not certify a tracking failure. That is why the score should be audited. It is not yet an account of where correspondence is lost.

4 Study 2: Measurement and selection audit

If the score does not fully reflect the trajectory, does it at least do the job people use it for—selecting among agents?

Three agents each ran a pre-registered schedule of 57 legs under one instrument: GPT-5.5, Qwen3.8-Flash, and Claude Opus 4.6. Tasks, interventions, seed, step budget, the inclusion threshold $n_{\min} = 3$ valid pairs to be ranked, and the rule that fewer than three ranked agents means the confirmatory selection test is *not evaluated* were committed before any outcome existed. A fourth lane was excluded before execution for instrument and transport mismatch and is not in Table 2. All three reported lanes used the same generic XML tool protocol. That is not the vendors’ native computer-use stack: comparability is within this roster and instrument only.

Coverage is the first result. Claude terminated on four legs across three tasks, so $|\mathcal{A}| = 1$ of 19 possible pairs. Under $n_{\min} = 3$ it is reported and not ranked. The ranked roster is two agents, which is fewer than three, and the confirmatory selection test is therefore not evaluated—a stated outcome of the plan, not a failed experiment. We do not compare models on 57 tasks. After the coverage-first rule, four task clusters support the remaining pairwise comparison.

Table 3: Common support. Rows are counterfactual-f010, preference_inference-f014, retrieval-f002, retrieval-f009. Δ : Flash–GPT.

Task	S^0 G/F	STS G/F	ΔS^0	Δ STS
cf-f010	83/100	0/0	+17	0
pref-f014	0/84	0/0	+84	0
retr-f002	75/100	0.50/0.50	+25	0
retr-f009	40/100	0.50/0.33	+60	-0.167

The measurement procedure itself constrains which comparisons can be made. Claude’s mean $S^0 = 100$ on $\mathcal{A}$ is a coverage artefact: it is the one task the agent solved and closed.

The intended tracking construct is degenerate. Binary track $Y = 1$ requires every positive-weight component of D to match on both legs. $Y = 0$ on all 18 valid pairs. There is no positive class, so S^0 cannot be calibrated to $P(Y = 1)$. Graded STS—typed extraction from the final answer against guest gold, frozen before any aggregate was computed, no judge in the loop—sits near its floor (Table 2). A later gap of a few hundredths is a gap between residuals, not between a reliable agent and an unreliable one.

Exploratory two-layer comparison. The plan permits one form of two-agent comparison: signed differences on the common support, labelled exploratory. $\mathcal{A}^\cap$ has four tasks. Differences are Flash minus GPT-5.5. A paired bootstrap over tasks (seed 20260904, $B = 5000$) gives $\Delta S^0 = +46.5$ (95% CI [21.0, 72.0]) and Δ STS = -0.042 (95% CI [-0.125, 0.000]): arg max S^0 is Flash and arg max STS is GPT-5.5.

The reversal is strict on one task and tied on three (Table 3). Removing retr-f009 makes the STS comparison a tie. A leave-one-pair-out check reports the same single-task concentration and is not a second confirmation. On these pairs STS takes only the values {0, 1/3, 1/2}. The disagreement sits at the resolution limit of the tracking layer.

Restricting to $\mathcal{A}^\cap$ drops GPT-5.5’s mean S^0 from 69.3 on its own nine pairs to 49.5, while Flash is unchanged (95.8 $\rightarrow$ 96.0). On each agent’s own $\mathcal{A}$, Flash is higher on both layers (STS 0.229 vs. 0.130). Which agent “wins” depends on the layer and on which tasks each agent finished. Neither denominator was chosen after seeing which looked better.

Conditioning on termination also changes what the score can see. Excluded cells score *lower* on average (mean S off $\mathcal{A}$: 46.4, 44.9, 26.1 against 69.3, 95.8, 100 on $\mathcal{A}$), so the filter is conservative for calibration rather than flattering. What survives that average is sharp: 2, 2, and 1 excluded cells scored ≥ 90 with no canonical DONE. Because termination never consults the judge, these are rubric-visible progress without closure, not scoring accidents. A completion-conditioned protocol cannot express them. A parser canonicalisation enumerated from one agent’s dialect is more forgiving of that agent: treating those rescues as rejections moves Flash from $|\mathcal{A}| = 8$ to 7 and leaves the other lanes unchanged. Reported $|\mathcal{A}|$ are lower bounds for that reason as well.

Table 4: Cause of the 39 recall misses. Meanings are locked; M2 is not a scope failure.

Cause	n	Meaning
M1a	13	gold entered found and was discarded
M1b	7	candidates disagree; gold is not among them
M2	9	no label hit anywhere in the answer
M3	5	label hit; gold outside every $\pm$ window
M4	5	instrument reported a non-gold value

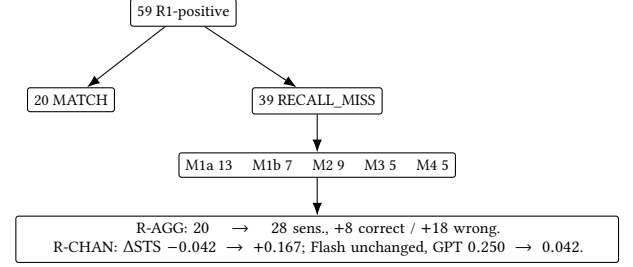


Figure 2: Study 3 hero: recoverable evidence is often missed; one mechanism is aggregation discard; repairs are signed.

5 Study 3: Diagnosing evidence loss

Studies 1 and 2 show that correspondence can fail. They do not say where. We freeze the evaluator and inspect the maps that turn a trajectory into a reported value.

The unit is a row: one component of D on one leg. The development archive contributes 134 rows over 57 legs. A row is *R1-positive* if the gold value is literally present in the agent’s answer ($n = 59$: 20 MATCH + 39 RECALL_MISS). The remaining 61 rows are ABSENT: gold is not in the text. All rates below are conditional on these audited sets. They are not a false-negative rate of the agent or of the benchmark.

Recall loss. The frozen instrument recovers 20 of 59 R1-positive rows (sensitivity 0.339). It misses 39. That is the headline of this study: recoverable information can be lost before scoring, so a score can fail to correspond to what happened.

The 39 misses are not one mechanism (Table 4, Figure 2). The strongest specific finding is M1a: in 13 of 39 misses the extractor had already accumulated the gold value and fail-closed aggregation discarded it. In 7 of those 13, gold was a strict majority of the accumulated candidates. M1 is not exclusive to misses (25 of 61 ABSENT rows are also M1); M1a is the cause that names recoverable-but-discarded evidence.

Repairs test mechanisms, not a method. Four layer interventions were specified before the run: R-AGG (plurality over accumulated candidates; targets M1a), R-SCOPE (whole-answer window; targets M3), R-CMP (containment comparison; targets entity M1b), and R-CHAN (delete markup spans; targets scaffolding). ALL composes them. No configuration is offered as a replacement instrument.

No repair dominates (Table 5). R-AGG is the only intervention that raises sensitivity (20 $\rightarrow$ 28). It releases 26 abstentions: +8 correct and +18 wrong. Most of the new errors are ABSENT rows

Table 5: Layer repairs as diagnostic interventions, 134 rows. Sensitivity is over 59 R1-positive rows. R-CMP is bit-identical to FROZEN.

Config	Sens./59	Abst./134	M4/reported	MATCH dest./20
FROZEN	20	89	25/45	0
R-AGG	28	63	43/71	0
R-SCOPE	13	56	65/78	14
R-CMP	20	89	25/45	0
R-CHAN	15	104	15/30	7
ALL	10	79	45/55	16

on which gold is not in the text at all—the instrument now commits where it previously stayed silent. Recovering more evidence is not equivalent to recovering the intended measurement.

R-SCOPE converts 14 of 20 already-correct rows into something else: raising observation recall *lowers* measurement correctness (confident-wrong 0.833). R-CHAN destroys 7 MATCH rows, mostly by deleting unterminated markup to the end of the text—a harm direction the fixtures already named. ALL is worse than every component on sensitivity. The four interventions do not vote. They diagnose different maps, and the maps do not move the measurement in one direction. R-CMP is bit-identical to FROZEN on every quantity: the comparison layer cannot reach an abstention, and the entity failures in this archive are abstentions. A later-layer comparison repair cannot recover an upstream miss.

Signed effects on a comparative statistic. On the same four-task support as Study 2, Flash-minus-GPT graded Δ STS is -0.0417 under FROZEN. R-AGG narrows the mean gap to -0.0208 while per-task disagreement grows (the two nonzero tasks move in opposite directions and partly cancel). R-CHAN inverts the sign: Δ STS = $+0.1667$, arg max moves to Flash. Under R-CHAN, Flash is bit-identical to FROZEN on every task; GPT collapses from 0.250 to 0.042. The ordering moved because the channel intervention removed one lane’s evidence, not because the other lane measured better. ALL is that destruction plus a Flash gain from the other repairs ($\Delta = +0.2292$).

This is an existence demonstration. The effective number of independent task-clusters for the frozen Δ STS is one (retr-f009). We make no claim about the prevalence of ranking instability. Downstream agreement with the rubric score is not evidence of measurement validity: R-CHAN agrees with S^0 by destroying GPT.

Specification portability. Separately, we ask whether the frozen instrument’s hand-written label table can be reconstructed by a corpus-independent, task-grounded rule R , locked before validation. Groundedness against the task definition is 100% (G1). Calibration fidelity against the frozen table is 9/30 (G2 FAIL). MATCH rows fall 20 $\rightarrow$ 13; ABSENT is unmoved. We did not ask whether R was better. We asked whether a pre-specified grounded rule could reconstruct the existing instrument. It could not. The 9/30 and the 39/59 are different estimands; they are not pooled.

A pre-specified comparative cohort gate required ≥ 20 independent task clusters under the same construct and frozen evaluator. A frozen feasibility probe—no agent, no tracking—passed 16 of 28 candidates (Wave A 9/12, Wave B 7/16). The comparative experiment was not opened.

Table 6: One constructed measurement, five correspondence checks. Populations are not added.

Stage	Intended correspondence	Observed failure
Trajectory $\rightarrow$ score	Score tracks outcome	Study 1: Type A/B dissociation
Evidence $\rightarrow$ selection	Score supports intended selection	Study 2: coverage; $Y=0$ on 18/18
Trajectory $\rightarrow$ evidence	Recoverable evidence survives	Study 3: 39/59 misses
Evidence $\rightarrow$ score	A layer fix preserves meaning	Study 3: +8 correct / +18 wrong; no dominance
Specification $\rightarrow$ instrument	Frozen labels are reconstructible	Study 3: 9/30

6 Unified findings

Table 6 is the paper. The score is not a direct observation of agent reliability. It is produced by transformations, and each transformation can introduce a correspondence failure. Study 1 detects the failure at the composite. Study 2 shows that the intended selection construct is coverage-limited and, in its binary form, degenerate. Study 3 locates losses inside the evaluator and shows that intervening on a layer can move both row-level decisions and, on the available support, a comparative statistic—without thereby validating the intervention.

The three studies share an instrument family and overlapping models. That is the design: one measurement system, three resolutions. It is not three corroborating experiments.

For agent evaluation the implication is operational. A CUA leaderboard that reports only S cannot distinguish a world-faithful agent from a well-scoring one (Study 1), cannot say which agents were even eligible to be compared (Study 2), and cannot say whether the number was produced by discarding evidence the agent already wrote (Study 3). The next instrument should expose those maps: a fail-closed termination channel disjoint from the judge; a tracking construct that does not consult the judge; an extractor frozen before aggregates exist; and a reconstruction test for any hand-written label table. None of those requirements is a new benchmark. They are conditions under which a reliability score can be interpreted as a measurement of the intended construct.

7 Related work

CUA benchmarks score completion [3, 4, 9, 10]. The intended reading is that a high number means the agent did the task in the world it was given. That reading is a construct-validity claim [2, 5]: the score is offered as a measurement of reliability, not as a record that a rubric fired. We take the claim literally and test the maps that produce the number.

Three nearby lines do not replace that test. Protocol-validity work asks whether the intended capability remains necessary for a score [7]. Counterfactual clinical evaluation shows that coverage-style scores can hide update failures under controlled mutations [8]. Judge-sensitivity benchmarks vary the prompt and hold the world fixed [1]. We hold the judge fixed and vary the world (Study 1), the analysis support (Study 2), and the internal maps of the instrument

(Study 3). The paired design uses interventionist language [6]; we identify no model internals.

The difference that matters for agent evaluation is the unit. Prior CUA leaderboards report a score per episode. We report a score against a world that was moved on purpose, a selection decision against a tracking construct that never consults the judge, and an evaluator against the evidence that was already in the answer. Those are three resolutions of one measurement, not three new benchmarks.

8 Limitations

One benchmark, one frozen instrument, a shared generic tool protocol rather than vendor-native computer-use stacks. Study 1’s invariance intervals are wide; Study 2’s confirmatory test was not evaluated; binary tracking has no positive instance; the exploratory reversal and the R-CHAN inversion rest on the same four-task support and the same single effective cluster. Repairs were a pre-specified family, not a search. R is one locked rule family; 9/30 is not a verdict on all hand-written instruments. Per-step guest state was not retained. We estimate no population rate.

9 Conclusion

A CUA reliability score is a constructed measurement. On this instrument, the number can come apart from the world, fail as a selection device, lose recoverable evidence, respond non-monotonically to layer interventions, and resist grounded reconstruction. The constructive implication is for the next measurement: name the transformations, freeze them, and test correspondence at each map—rather than reading reliability off the scalar they emit.

AI use

Generative tools were used to draft and copy-edit prose against a frozen claim-to-evidence lock. No tool chose a hypothesis, a gate, a repair, or a number. Every statistic was checked against the recorded analysis artefacts. The authors take responsibility for the text.

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